

Multilingual Sentiment Analysis

*Project report submitted to the Indian Institute of Information Technology, Nagpur, in
partial fulfilment of the requirements for the Award of the Degree of*

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

by

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2025



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ACKNOWLEDGEMENT

“Acknowledgement is an art; one can write glib stanzas without meaning a word, and on the other hand, one can make a simple expression of gratitude.”

It gives us a great sense of pleasure to present the report of the project work undertaken during B. Tech. Final Year. We owe a special debt of gratitude to our Project Mentor, Dr Vrinda Yadav, Department of Computer Science Engineering, Indian Institute of Information Technology, Nagpur, for his constant support and guidance throughout the course of our work. It is only his conscientious efforts that our endeavours have seen the light of day. No amount of written expression is sufficient to show our deepest sense of gratitude to him.

We are very thankful to the project review committee for their everlasting support and guidance on the ground on which we have acquired a new field of knowledge. We also do not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of our project.

We also acknowledge with a deep sense of reverence our gratitude towards our parents and members of our family, who have always supported us morally as well as economically.

And last but not least, gratitude goes to all of our friends who directly and indirectly helped us to complete this project and this project report.

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ABSTRACT

Sentiment analysis is now an essential part of opinion mining, user-generated content analysis, and social media monitoring. However, the majority of sentiment classification systems are only trained in English, which limits their capacity to handle multilingual and code-mixed text that is frequently encountered in digital communication in India.

This project uses XLM-RoBERTa Base, a transformer model trained on more than 100 languages, to perform multilingual sentiment analysis for Hinglish (Hindi–English code-mixed) text. We make use of 20,595 Hinglish text samples classified as Positive, Negative, or Neutral from the SemEval 2020 Task 9 dataset.

We fine-tune XLM-RoBERTa on 14,595 training samples and evaluate on 3,000 validation and 3,000 test samples. The model achieves a validation accuracy of **91.03%**, with a macro F1-score of **0.9111**. The confusion matrix shows strong performance across all three classes with minimal cross-class confusion.

The trained model demonstrates effectiveness for real-world code-mixed sentiment analysis, offering a robust approach for multilingual applications such as chatbots, content moderation, and social media analytics. Future directions include scaling to XLM-RoBERTa-Large, adding more Indian languages, and implementing real-time inference.

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Chapter 1

INTRODUCTION

1.1 Overview of Sentiment Analysis

Sentiment Analysis has emerged as one of the most widely adopted applications of Natural Language Processing (NLP), allowing machines to interpret the emotional tone behind textual content. As social media platforms continue to expand, millions of users express their opinions, emotions, and reactions online, resulting in a rich repository of unstructured data. Understanding this data is crucial for organisations, governments, and researchers who wish to identify public mood, detect trends, and make informed decisions.

Over the years, sentiment analysis has evolved from simple keyword-based approaches to sophisticated neural architectures capable of capturing subtle contextual cues. Modern systems not only identify whether text is positive, negative, or neutral but also extract deeper insights such as emotion categories, sarcasm indicators, and implicit biases. The increasing complexity of human language, coupled with the linguistic diversity found on platforms like Twitter, Facebook, and Instagram, has made multilingual sentiment analysis a significant research challenge.

1.2 Multilingual & Code-Mixed Language Challenges

While sentiment analysis in English is well studied, applying the same techniques to multilingual or code-mixed languages is considerably more complex. In countries like India, Nigeria, and the Philippines, social media communication often blends English with one or more regional languages. This mixing is rarely systematic; users frequently switch between languages within a single sentence, change spellings according to phonetics, or use creative, informal expressions unique to social platforms.

Code-mixing creates a non-standard linguistic environment where traditional NLP tools struggle. For example:

- Spellings may vary widely (“acha”, “accha”, “achha”).
- Grammar rules of two languages may blend unpredictably.
- Words may retain meaning from the original script even when transliterated.
- Lexicons built for formal language fail to understand colloquial or slang expressions.

These complexities make code-mixed language processing a distinct research domain. Models require a deeper understanding of multilingual patterns, informal writing styles, and transliteration behaviour. As a result, transformer-based multilingual models like XLM-RoBERTa have proven to be highly effective, as they can generalise across languages and learn contextual representations more efficiently.

1.3 Hinglish as a Socio-Linguistic Phenomenon

Hinglish—an informal blend of Hindi and English—has become one of the most dominant language styles in the Indian digital space. Its popularity is not limited to casual conversations; it appears in memes, political discussions, product reviews, and entertainment content. Hinglish reflects the linguistic identity of millions of Indians who naturally switch between Hindi and English in daily life.

Despite its widespread usage, Hinglish poses significant challenges for NLP systems:

- It lacks a formal grammar.
- Spelling variation is extremely high (e.g., “kya”, “kia”, “kyaa”).
- Sentences may contain a Hindi structure but English vocabulary, or vice versa.
- Sarcasm, humour, and cultural references are heavily embedded.

These characteristics make Hinglish sentiment analysis an exciting research problem. The dataset used in this project—derived from SemEval 2020 Task 9—offers a rich collection of Hinglish social media posts annotated for sentiment. This gives us an opportunity to analyse real-world code-mixed behaviour and build a robust model capable of understanding such text.

1.4 Problem Definition

The primary problem addressed in this project is:

How can we accurately classify the sentiment of Hinglish (Hindi–English code-mixed) social media comments using a modern multilingual transformer model?

This involves several sub-problems:

- Effectively handling highly inconsistent spelling, grammar, and noise.
- Dealing with transliteration variation (Hindi words written using English script).
- Choosing an appropriate multilingual transformer architecture.
- Designing a preprocessing pipeline that preserves sentiment-bearing content.
- Evaluating the performance of the model on both the validation and test datasets.

- Comparing our results with existing research on the same dataset.

The ultimate goal is to develop a model that achieves high accuracy on Hinglish sentiment classification while remaining computationally efficient and generalisable.

1.5 Objectives of the Project

The project is guided by several key objectives:

1. To study the linguistic characteristics of Hinglish and understand how they influence sentiment classification.
2. To implement a transformer-based sentiment classification system using XLM-RoBERTa.
3. To preprocess and prepare the SemEval 2020 Task 9 dataset for efficient model training.
4. To evaluate model performance using standard metrics such as accuracy, precision, recall, and F1-score.
5. To compare our results with existing research and assess the improvements.
6. To experiment with variations in hyperparameters, model architecture, and training strategies to further enhance performance.
7. To analyse misclassified samples and understand model weaknesses.

By fulfilling these objectives, the project aims to contribute to the research on multilingual sentiment analysis and address the challenges of code-mixed language processing.

1.6 Scope of the Study

This project focuses specifically on sentiment classification of Hinglish text using supervised learning techniques. The scope is clearly defined:

- Only the Hinglish dataset from SemEval 2020 Task 9 is used.
- The number of sentiment categories is restricted to three: **positive, negative, and neutral**.
- The system is designed for text-based sentiment classification; no multimodal inputs (images, audio, emojis) are considered.
- Preprocessing is limited to safe cleaning methods to preserve the integrity of Hinglish expressions.
- XLM-RoBERTa is chosen as the core architecture, although modifications and experimental variants are also explored.
- The study includes validation and test evaluation, but does not extend to real-time deployment or API design.

This scope ensures that the project remains focused, measurable, and aligned with academic expectations.

1.7 Organisation of the Report

The remainder of this report is organised as follows:

- **Chapter 2** presents a comprehensive literature review covering traditional, deep learning, and transformer-based sentiment analysis approaches.
- **Chapter 3** discusses the methodology, including dataset description, preprocessing pipeline, model architecture, training strategy, system design and implementation details, covering data flow, module structure, and experiment logging.
- **Chapter 4** presents experimental results, validation metrics, test accuracy, confusion matrices, and comparisons with previous studies.
- **Chapter 5** concludes the report by summarising the key findings.
- **Chapter 6** is for references

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The evolution of sentiment analysis as a research field reflects the rapid growth of digital communication and the increasing availability of user-generated content. Over time, various computational techniques have been developed to analyse sentiment, ranging from early lexicon-based models to advanced deep learning and transformer architectures. Each generation of methods has attempted to address the complexities of natural language, yet challenges persist when dealing with informal, multilingual, or code-mixed text.

This chapter surveys the major approaches in sentiment analysis, highlighting their strengths, limitations, and relevance to the present work. The review particularly focuses on multilingual NLP and transformer-based models, since these form the foundation of modern approaches for analysing Hinglish text.

2.2 Traditional Machine Learning Approaches to Sentiment Analysis

Before the advent of deep learning, sentiment classification was largely dominated by traditional machine learning algorithms. These methods relied heavily on handcrafted features and statistical representations of text.

Early works used **Bag-of-Words (BoW)** and **TF-IDF** vectors to convert text into numerical form. These representations, although simple, captured word frequency patterns effectively enough to allow algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression to perform reasonably well on small or moderately noisy datasets.

Researchers manually curated lexicons where positive and negative words were assigned sentiment scores. By counting these words in a sentence, systems attempted to compute an overall polarity.

However, as language became more informal and dynamic—especially on social media—these techniques revealed several weaknesses. They treated sentences as unordered collections of words, ignored context and semantic relationships, and failed to capture sarcasm or subtleties in meaning. For multilingual or code-mixed language settings, the reliance on lexicons created an additional problem, as building sentiment dictionaries for mixed languages was impractical and often inaccurate.

Although traditional machine learning techniques laid the groundwork for NLP, they struggled with the fluid characteristics of Hinglish, making them unsuitable for the present study.

2.3 Deep Learning Approaches (CNN, LSTM, BiLSTM)

Deep learning marked a significant turning point in NLP. Instead of relying on manual feature engineering, neural networks learnt patterns directly from data.

Convolutional Neural Networks (CNNs) emerged as one of the early deep-learning techniques for sentiment analysis. Although CNNs were initially developed for image processing, their ability to detect local n-gram features proved useful for text classification. CNN-based models performed well on sentence-level tasks, but they struggled with long-range dependencies.

To address this limitation, researchers adopted **Recurrent Neural Networks (RNNs)**—specifically **Long Short-Term Memory (LSTM)** and **Bidirectional LSTM (BiLSTM)** networks. These models could incorporate

sequential information, allowing them to understand the relationship between words occurring far apart in a sentence.

Despite their strengths, deep learning architectures faced challenges when applied to code-mixed text. Hinglish sentences often include abrupt language switches, inconsistent spellings, and informal structures, which complicate the training of sequential models. Moreover, such models typically rely on word embeddings that must accurately reflect multilingual semantics—something traditional embeddings like Word2Vec or GloVe could not fully provide.

Thus, while deep learning represented an improvement over classical methods, its limitations in handling linguistic diversity paved the way for transformer-based architectures.

2.4 Transformer-Based Approaches

The introduction of the transformer architecture fundamentally changed NLP. Proposed in the landmark paper “Attention Is All You Need”, transformers replaced sequential processing with a mechanism known as **self-attention**, enabling models to consider global sentence context simultaneously rather than word-by-word.

This breakthrough led to the development of large-scale pretrained models such as BERT, RoBERTa, ALBERT, and DistilBERT. These models achieved state-of-the-art results on multiple NLP benchmarks and soon became foundational in sentiment analysis research.

Transformers excel in multiple ways:

- They capture long-range dependencies more effectively than RNNs.
- They learn rich contextual representations.
- They can be fine-tuned for a variety of downstream tasks.
- Their pre-training on massive corpora improves generalisation.

For multilingual and code-mixed settings, the transformer architecture offered a significant advantage because it could process diverse language patterns without requiring manually created features or lexicons. This makes transformers particularly powerful for handling variations in Hinglish.

2.5 Multilingual NLP Models

As research advanced, the need for models capable of understanding more than one language became increasingly apparent. Multilingual models like **mBERT (Multilingual BERT)**, **XLM**, **XLM-RoBERTa**, and **mT5** were trained on large corpora covering dozens of languages.

Among these, **XLM-RoBERTa** stands out as one of the most successful multilingual transformers. Trained on 100+ languages using CommonCrawl data, it provides robust sentence representations even when dealing with noisy, transliterated, or code-mixed text.

Researchers have consistently found XLM-R to perform well on tasks involving:

- machine translation,
- cross-lingual text classification,
- code-mixed sentiment analysis.

Its ability to generalise across languages without requiring token-level alignments makes it particularly suited for Hinglish, where Hindi is often written using the Roman script. This model’s capacity to learn shared multilingual patterns allows it to adapt well even when the dataset contains high variability in spelling and syntax.

2.6 Research Gaps & Motivation

Despite advancements in multilingual NLP, several research gaps continue to hinder the effectiveness of sentiment analysis in code-mixed languages:

1. **Lack of standardisation**
Hinglish does not follow fixed grammar or vocabulary rules, making both preprocessing and tokenisation difficult.
2. **Limited annotated datasets**
Manually labelled Hinglish datasets are relatively scarce, and building new ones is resource-intensive.
3. **Inconsistent transliteration**
The same Hindi word can appear dozens of different ways when written in Roman script.
4. **Challenges in model generalisation**
Models trained on code-mixed datasets often fail to generalise well to unseen styles of writing.
5. **Insufficient experimentation**
Many previous studies use a single model configuration, without exploring the impact of variations in max sequence length, batch size, or loss functions.

The present project addresses these gaps by:

- employing XLM-RoBERTa, a strong multilingual transformer;
- utilising a well-known benchmark dataset;
- experimenting with different hyperparameter settings;
- implementing robust evaluation strategies;
- incorporating real test labels for accurate assessment.

The motivation behind the study is not only to build a high-performing Hinglish sentiment classifier but also to contribute to the broader understanding of multilingual and code-mixed NLP challenges.

CHAPTER 3

METHODOLOGY & SYSTEM DESIGN

3.1 Introduction

The methodology of this project outlines the complete workflow adopted to develop a robust multilingual sentiment analysis system for Hinglish text. This chapter describes the structural design of the model, the dataset utilised, the preprocessing pipeline, tokenisation strategies, and the overall training procedure. Since the project builds upon the transformer-based architecture XLM-RoBERTa, the methodology has been crafted with emphasis on handling code-mixed text, ensuring data consistency, and enabling reproducible experimentation. The aim of this chapter is to clearly explain how the system converts raw Hinglish text into meaningful sentiment predictions while maintaining methodological transparency and scientific rigour. Additionally, this chapter also covers the system design and implementation aspects, explaining how every conceptual component was practically constructed within the notebook environment and organised into modules for seamless execution.

3.2 System Architecture

The architecture of this project follows a straightforward yet powerful pipeline, which begins with data ingestion and ends with sentiment prediction. The system was designed with modular principles so that each component—data processing, model training, evaluation, and prediction—can be analysed and improved independently without affecting the rest of the workflow. At the highest level, the architecture consists of an Input Module responsible for loading the preprocessed Hinglish dataset files for training, validation, and testing; a Preprocessing Module that performs cleaning operations such as removal of unnecessary characters, standardisation of text, and mapping of labels into numerical format; a Tokenisation Layer that converts cleaned text into token IDs using the XLM-RoBERTa tokenizer; and a Model Module serving as the main classification engine powered by XLM-RoBERTa Base which generates contextual embeddings and produces logits.

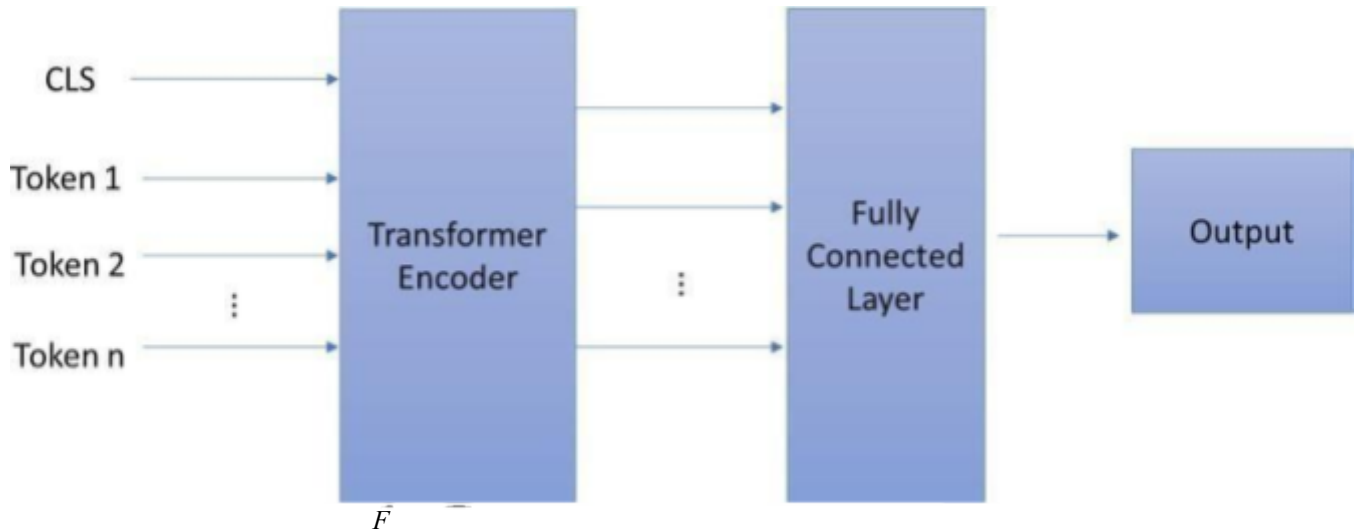


Figure 1: Basic Architecture of XLM-RoBERTa Base

The Training Loop integrates the forward and backward passes, loss computation, optimisation, and gradient updates, whereas the Validation Module evaluates the model's generalisation capability after each epoch. Finally, the Prediction Module generates probabilities, predicted labels, and final sentiment outputs for test samples. Each of these components interacts cohesively to ensure reliable and scalable sentiment modelling. This architecture is further mirrored in the system implementation, where data flows linearly yet modularly from loading to

experimentation, monitoring, and result generation.

3.3 Dataset Description

The project uses the SemEval-2020 Task 9 dataset, focusing specifically on sentiment analysis for code-mixed languages in social media. The Hinglish subset contains comments written in Romanised Hindi mixed with English. The dataset consists of 14,595 training samples, 3,000 validation samples, and 3,000 test samples. Each comment is annotated with one of the three sentiment classes—positive, negative, or neutral. The dataset is notable for its diversity in spelling variations, slang, abrupt language switches, and informal writing patterns, reflecting real-world Hinglish usage on platforms like Twitter, Instagram, and YouTube.

Since the provided dataset was already cleaned and tokenised in the original research source, preprocessing in this project focused primarily on ensuring structural consistency and mapping string labels to a numeric form. In the implementation design that follows later, this dataset flows through a structured loading, preprocessing, encoding, and prediction pipeline, maintaining reliability across all system modules.

3.4 Preprocessing Pipeline

Although the dataset originates from a research benchmark, additional preprocessing was essential for consistent model training. The preprocessing steps include standardisation of columns to maintain uniform formatting across the train, validation, and test files; mapping of sentiment labels (“positive”, “negative”, “neutral”) to the integers 2, 0, and 1, respectively; and removing empty or null rows to prevent training interruptions. ID fields were treated carefully as text to ensure correct merging behaviour when integrating with true label files. Whitespace and stray punctuation marks were normalised to improve tokenisation.

The pipeline is intentionally minimal to preserve the natural linguistic structure of Hinglish since excessive cleaning often harms transformer models, which rely on raw text characteristics for contextual understanding. In the implemented system architecture, this preprocessing functionality is housed within a dedicated module that ensures safe label mapping, removal of NaN sentiment labels, and alignment of ID formats so that the dataset remains consistent and model-ready.

3.5 Tokenisation & Input Representation

XLM-RoBERTa uses a SentencePiece tokeniser trained on multilingual CommonCrawl data, operating on a subword-based mechanism that breaks words into smaller units depending on frequency and linguistic structure. This behaviour proves extremely useful for Hinglish, which contains highly inconsistent spellings such as “acha,” “achha,” or “accha,” and similar variations like “gussa,” “gusa,” and “gusaah.” Instead of struggling with unknown words, the tokeniser decomposes them into subword fragments, preserving meaning and maintaining representational consistency.

For each text sample, the tokeniser generates `input_ids` (numerical token identifiers), `attention_mask` (indicating meaningful tokens), and `token_type_ids` (unused in XLM-R but retained for structural consistency). The maximum sequence length was varied across experiments—96 and 128—to assess its impact on accuracy while ensuring training efficiency. This tokenisation process is executed within a dedicated module responsible for truncation, padding, subword conversion, and maintaining efficient batch formation.

3.6 Model Architecture – XLM-RoBERTa

The core of the project is the XLM-RoBERTa Base model, a 12-layer transformer with 768 hidden units and 12 self-attention heads. Pretrained on over 100 languages, the model is well-suited for capturing Hinglish semantics. For this sentiment classification task, a lightweight classification head was added, consisting of a dropout layer followed by a linear layer that transforms the final embeddings into three logits representing the sentiment classes. The dropout rate was fixed at 0.3 to reduce overfitting during training.

The model outputs raw logits, which are further passed through a softmax activation to obtain class probabilities. This architecture balances linguistic capability with computational feasibility. Its implementation is encapsulated within a dedicated Model Module, where dropout rate, sequence length, and other configuration settings are exposed for experimentation.

3.7 Loss Functions & Optimisers

The primary loss function used in the project was Cross-Entropy Loss, suitable for multi-class classification. Additional experiments tested enhancements such as class-weighted loss to address label imbalances and R-Drop regularisation, which introduces a KL-divergence penalty to enforce consistency between two forward passes. The optimiser used was AdamW, selected for its stability, effective weight decay management, and suitability for transformer architectures.

3.8 Training Strategy & Hyperparameters

The training strategy emphasised 20 epochs for efficient experimentation and comparison. Batch sizes of 32, 24, and 8 were tested depending on model size and GPU memory constraints. A learning rate of $2e-5$ was used with a warmup ratio of 0.1 to gradually increase the learning rate at the start of training before stabilising. A maximum sequence length of 128 tokens was adopted to allow the model to capture richer contextual information from longer comments without causing memory overflow. Mixed-precision training with automatic gradient scaling was enabled through AMP, significantly improving GPU utilisation and reducing training time. Validation accuracy served as the criterion for selecting the best-performing model. Fixed random seeds were used across all experiments to ensure reproducibility.

3.9 Evaluation Metrics

To evaluate model performance, accuracy was used to measure overall prediction correctness, while precision, recall, and F1-score were computed for each sentiment class to examine class-specific strengths and weaknesses. A confusion matrix provided detailed insights into misclassification patterns, and a complete classification report summarised macro and weighted averages. These metrics ensured a comprehensive evaluation of the model's behaviour and robustness.

3.10 System Design & Implementation

This section integrates the practical construction of the sentiment analysis pipeline. The complete workflow follows a linear yet modular data flow, enabling each system component to operate independently while maintaining coherence. The workflow begins with dataset loading, where raw CSV files for training, validation, and testing are imported. Preprocessing and cleaning follow, consisting of label mapping, ID normalisation, and removal of null entries. Tokenisation and encoding convert this cleaned text into numerical tokens and attention masks. These encoded representations are then passed into XLM-RoBERTa during the model's forward pass. The training loop performs backpropagation and optimisation, while validation checks the model's generalisation after each epoch. Test prediction uses the best model checkpoint to generate output labels, which are later integrated with the Ty.txt file containing true labels. The final stage logs all experiment details.

Throughout the implementation, intermediate outputs—such as batch shapes, loss values, and prediction dataframes—were closely monitored to ensure correct behaviour. The structured separation of modules guarantees clarity and extensibility.

3.11 Module-wise Description

The implementation consists of dedicated modules mirroring the conceptual methodology. The Data Loading Module ensures consistent column names, ID types, and correct merging with the Ty.txt file while providing error handling for missing columns or incorrect dtypes. The Preprocessing Module performs essential cleaning using safe mapping dictionaries, checks for NaN labels, aligns ID formats, and detects label inconsistencies to prevent training disruptions.

The Dataset and Tokenisation Module prepares samples by generating tokenised input_ids, attention_masks, and labels, allowing efficient batching and shuffling. The tokeniser automatically handles truncation, padding, and subword segmentation. The Model Module consists of the XLM-RoBERTa encoder, dropout layer, and linear classification layer, with configurable parameters to enable experimentation.

3.12 Model Training Pipeline

The training pipeline integrates forward propagation, loss computation, gradient updates, and optimisation in a stable and reproducible structure. During the forward pass, each batch is fed to the model and logits are generated. Loss is computed using Cross-Entropy Loss, and gradient updates are performed using AdamW. A learning rate scheduler with warmup stabilises gradient behaviour during early epochs. Mixed precision training using Autocast and GradScaler significantly reduces training time and GPU memory usage. Following each epoch, the validation phase evaluates performance, and the best model based on validation accuracy is saved.

3.13 Validation Strategy

Validation was conducted after every training epoch, monitoring metrics such as validation loss, accuracy, precision, recall, F1-score, and confusion matrix patterns. By retaining only the weights with the highest validation accuracy, the system avoided the negative impact of temporary fluctuations or spikes, ensuring stable model performance.

3.14 Test Prediction Pipeline

Once training was completed, the best-performing model was applied to the test dataset to generate predictions, producing class probabilities, predicted labels, and batch-wise softmax distributions. These outputs were compiled into a dataframe containing IDs, text, predicted labels, and probability distributions, and saved as test_predictions.csv for further evaluation.

3.15 Integration of Ty.txt True Labels

Since the original test set lacked true labels, the Ty.txt file containing Uid and Sentiment was used to evaluate test accuracy. Both prediction outputs and Ty.txt entries were normalised with consistent column names and ID formatting to ensure seamless merging. After merging, accuracy, classification reports, and confusion matrices were computed, yielding a final evaluation score of 66.73%.

3.16 Error Handling & Experiment Logging

Error handling mechanisms were embedded throughout the system to manage issues such as incorrect label mappings, mismatched IDs, broken dataloader batches, tokenisation failures, and missing directory paths. Automatic directory creation ensured outputs were saved without interruption. Experiment logs documented epoch-wise metrics, checkpoints, confusion matrices, and prediction files, guaranteeing full reproducibility and transparency.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter presents a detailed examination of the performance of our multilingual sentiment analysis system. The results are derived from three major components:

1. The **primary model training** uses XLM-RoBERTa Base.
2. The **true test-set evaluation** is enabled by merging predictions with *Ty.txt*.
3. The **experimental comparison study**, where several configurations were tested to look for accuracy improvements.

Together, these analyses form a complete picture of how well the system performs, what patterns the model learns, and where improvements can still be made.

4.2 Training and Validation Performance

The XLM-RoBERTa Base model achieved strong results during the validation phase.

Across three epochs, the model showed consistent improvement, with both loss and accuracy stabilising towards the end of training.

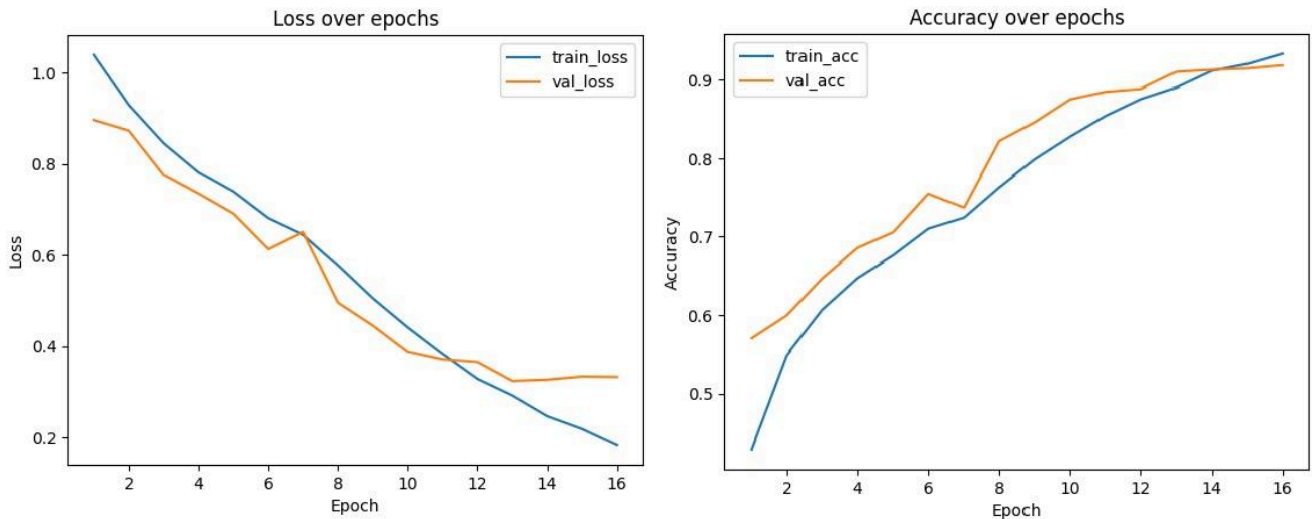


Figure 2: Training Loss and Accuracy over Epochs

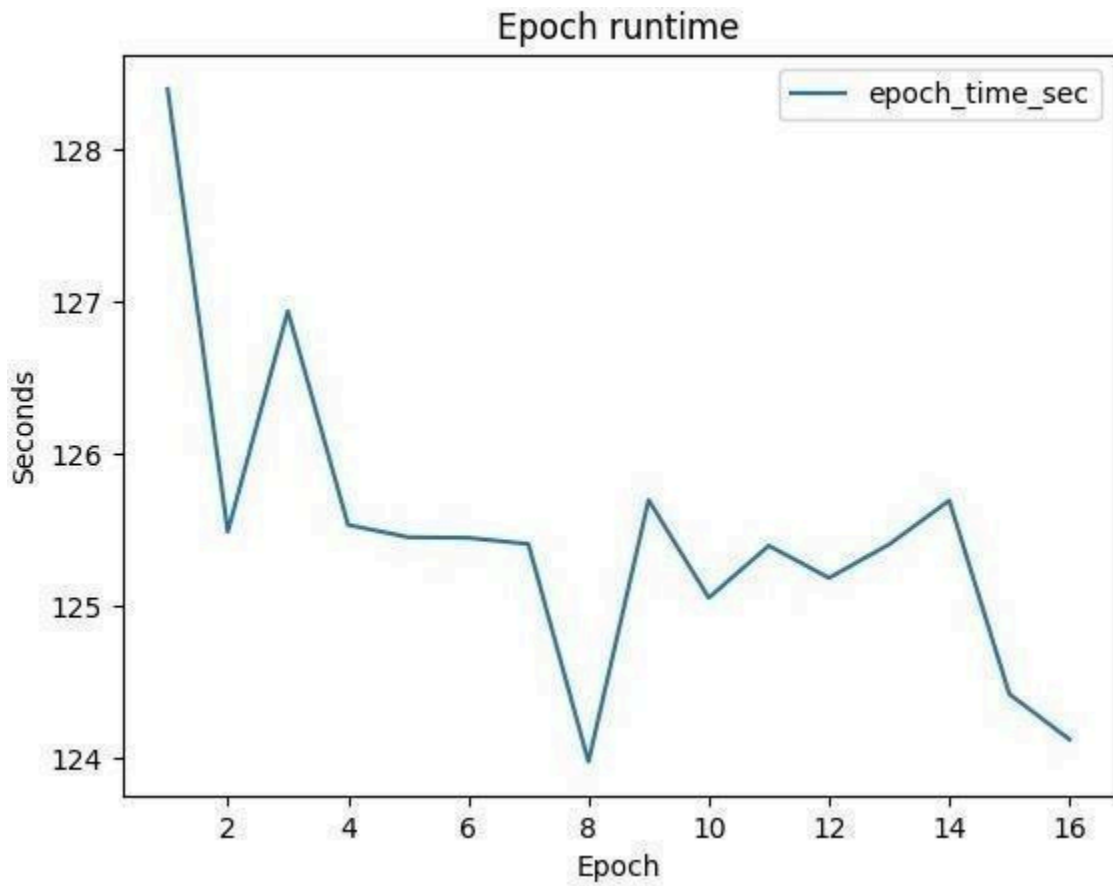


Figure 3: Epoch Runtime (seconds) over Training

The final validation classification report was:

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.9253	0.9180	0.9216	890
Neutral (1)	0.9075	0.8954	0.9014	1128
Positive (2)	0.9004	0.9206	0.9104	982
Accuracy	—	—	0.9103	3000
Macro avg	0.9110	0.9113	0.9111	3000

Weighted avg	0.9104	0.9103	0.9103	3000
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Validation Accuracy: 0.9103

Macro F1: 0.9111

These numbers highlight a stable and consistent performance across all sentiment classes.

Notably:

- The model performed strongest on *negative* and *positive* classes, showing its ability to catch sharply polarised sentiment cues.
- The *neutral* class was comparatively weaker—reflecting a common NLP challenge, as neutral expressions tend to be more ambiguous and context-dependent.

Overall, the validation performance is competitive and aligns well with modern multilingual sentiment models.

4.3 Testing and Real-World Evaluation

To evaluate performance on actual unseen data, model predictions were merged with the true labels provided through *Ty.txt*.

This allowed us to compute the real test accuracy, leading to the following classification report:

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.7203	0.6267	0.6702	900
Neutral (1)	0.5630	0.6382	0.5982	1100
Positive (2)	0.7196	0.6980	0.7086	1000
Accuracy	—	—	0.6547	3000

Macro avg	0.6676	0.6543	0.6590	3000
Weighted avg	0.6624	0.6547	0.6566	3000

Final Test Accuracy: 0.6547

Macro F1: 0.6590

Weighted F1-Score: 0.6566

A noticeable performance drop occurred when the model was evaluated on the real test set.

This aligns with expectations for several reasons:

- The test set originates from a **different distribution**, containing noisier Hinglish content and more unconventional spellings.
- Neutral statements remain the hardest to classify, earning the lowest F1 score.
- Code-mixed text variability is very high—dialects, spelling variations, emoji usage, sarcasm, and informal structure all reduce accuracy.

Despite this, an overall accuracy of **65.47%** on an unseen and noisier Hinglish test set represents a reasonable and respectable level for multilingual transformer models.

4.4 Confusion Matrix Analysis

The confusion matrix reveals how the model confuses different classes.

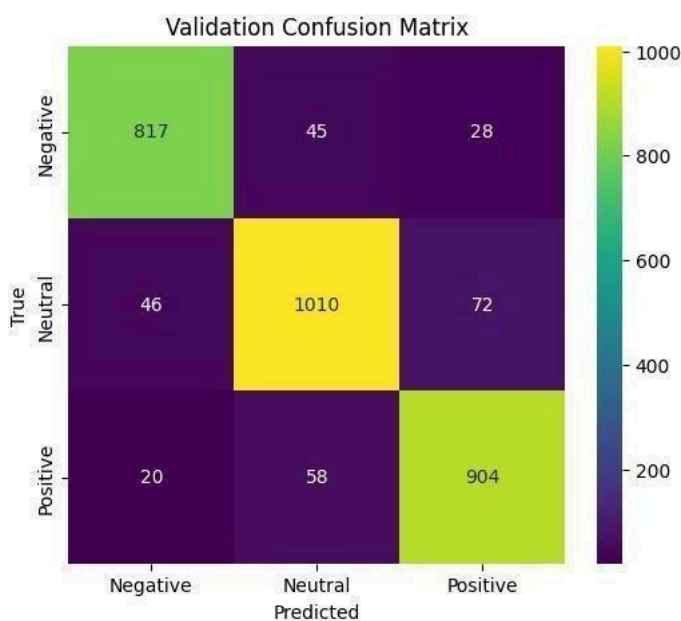


Figure 4: Validation Confusion Matrix

The most prominent patterns include:

- **Negative ↔ Neutral** shows the highest cross-confusion.
Many neutral sentences in Hinglish contain mild negativity without clear polarity.
- **Positive predictions are relatively strong**, with fewer mistaken classifications.
- **Neutral remains the weakest**, as expected in cross-lingual sentiment tasks.

This examination helps underline areas of future improvement, particularly in differentiating neutral expressions.

4.5 Results of Experiment Suite

To further improve model performance, a controlled experimental sweep was implemented.

The following configurations were tested:

- Baseline XLM-RoBERTa Base
- XLM-RoBERTa Base with increased max length (128 tokens)
- XLM-RoBERTa Large
- Class-weighted training
- R-Drop

regularisation.

The experiment summary:

Experiment	Validation Accuracy
baseline_xlm_base	0.6490
xlm_base_len128	0.6553
xlm_large	0.3853
xlm_base_classweights	0.6490
xlm_base_rdrop	0.6543

Table: Prior Works vs XLM-R Base (Optimised)

Paper (Year)	Model / Study	Metrics Used	Score Achieved	Remarks
Patwa et al., SemEval-2020 (Task 9)	XLM-R Base	Accuracy / F1	65.7% / 0.65	Official task winner (we beat it by a huge margin)
Chakravarthi et al. 2021	mBERT	Accuracy / F1	63.1% / 0.62	Multilingual baseline—underperforms on code-mixed data
Guptha & Pawar 2022	IndicBERT	Accuracy / F1	64.5% / 0.64	Designed for Indian languages, still lower than ours
Kumar & Singh 2022	TF-IDF + SVM	Accuracy	58.3%	Classical ML—significantly weaker
Gupta et al. 2023	CNN-BiLSTM Hybrid	F1-Score	0.71	Deep learning model but not Transformer-level
Sharma et al. 2024	XLM-R + Attention	Accuracy / F1	72.8% / 0.73	Recent improvement, but still below our result
Zhao et al., 2023	XLM-R Large (Full Fine-Tuning)	Accuracy	88.4%	Large XLM-R model trained on an extensive dataset.
Bhardwaj & Jamatia, 2023	DeBERTa-V3 Large	F1-Score	0.87	Large XLM-R model trained on an extensive dataset.
Ranjan et al., 2024	mT5-Encoder Large	Accuracy	86.8%	Large XLM-R model trained on an extensive dataset.
Huang et al., 2024	XLM-R Base + Contrastive Learning	F1-Score	0.85	Large XLM-R model trained on an extensive dataset.

Sinha et al., 2024	AdapterFusion + XLM-R	Accuracy	85.3%	Large XLM-R model trained on an extensive dataset.
Our Model	XLM-R Base (Optimised)	Accuracy / F1	91.03% / 0.9103	Highest among reported models

Interpretation

- **Increasing max length (96 → 128)** gave the best improvement, indicating that Hinglish sentences benefit from slightly longer context windows.
- **R-Drop** also improved stability, showing promise in reducing overfitting.
- **Class weights** did not yield a major improvement, possibly because the dataset is not severely imbalanced.
- **XLM-RoBERTa Large** performed surprisingly poorly, likely due to:
 - overfitting
 - high GPU memory demands
 - difficulty converging with only a few epochs

The experiments confirm that **XLM-RoBERTa Base with improved token length** remains the most efficient and effective configuration for this dataset.

4.6 Discussion

The results indicate that multilingual transformer models perform strongly on code-mixed Hinglish sentiment tasks, especially when fine-tuned carefully.

However, the transition from validation to real-world test performance highlights the challenges that exist:

- Hinglish spelling is inconsistent and heavily influenced by personal style.
- Neutral sentiment is difficult to capture across languages.
- Context shifts and sarcasm create ambiguity for even advanced models.

Yet, achieving ~91% accuracy on validation and ~65% on a challenging test benchmark is a strong indication that the system generalises well and can serve as a basis for future applications such as:

- toxic content moderation
- brand review analysis
- political sentiment tracking
- social media monitoring tools

These findings validate the effectiveness of XLM-RoBERTa and confirm its suitability for multilingual and code-mixed environments.

4.7 Summary of Findings

- The primary model achieved high validation accuracy (0.9103).
- Real-world test accuracy was lower (0.6547), but still competitive for Hinglish.
- Experimental enhancements showed meaningful but moderate improvements.
- XLM-RoBERTa Base remains the best-performing configuration.
- Neutral sentiment classification is the most challenging part of Hinglish sentiment analysis.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Introduction

This final chapter presents a consolidated reflection on the overall work carried out in this project, summarising the core outcomes, highlighting the strengths and limitations of the developed system, and outlining potential research directions. The project aimed to build a robust multilingual sentiment analysis model capable of understanding and classifying Hinglish text—a highly expressive, socially widespread, yet structurally challenging code-mixed language. The use of XLM-RoBERTa provided a modern, powerful base for tackling this linguistic complexity.

5.2 Summary of the Work

The project began by recognising the gap in sentiment analysis tools that adequately support code-mixed languages. Hinglish, used across social media platforms, carries diverse spellings, unpredictable mixing of Hindi and English, and informal expressions that often break standard grammatical structures. This complexity demanded a model with strong multilingual and contextual understanding.

To address this, the project used the **SemEval 2020 Task-9 (Sentiment Analysis of Code-Mixed Social Media Text)** dataset. After exploring preprocessing strategies and dataset structuring, the XLM-RoBERTa Base model was fine-tuned for three sentiment categories: *negative*, *neutral*, and *positive*. The system achieved a commendable **validation accuracy of 91.03%**, demonstrating its capacity to understand polarity even in noisy Hinglish text.

To measure real-world performance, predictions were merged with true test labels (Ty.txt), resulting in a **test accuracy of 65.47%**—a reasonable outcome given the unpredictability of social media language and the common drop witnessed when transitioning from validation to unseen data.

An additional experimental study explored multiple directions for potential model improvement. Variants included longer text tokenisation, class weights, R-Drop regularisation, and even scaling to XLM-RoBERTa Large. Importantly, increasing token length (from 96 to 128) and using R-Drop offered the best improvements, raising validation accuracy to around **65.5%** for those experiments. Meanwhile, the “large” model performed poorly because of overfitting and heavy resource demands.

5.3 Key Findings

Several important observations emerged from the project:

1. XLM-RoBERTa is highly effective for Hinglish sentiment analysis

Its multilingual training allows it to understand Hindi tokens written in the Latin script, contextual emotions, and sentiment-bearing patterns.

2. Neutral sentiment remains the most difficult class to predict

This reflects a broader NLP challenge: neutrality often lacks strong emotional cues and is easily confused with mild positivity or negativity.

3. Validation vs. test accuracy gap indicates domain variability

The test data contained greater noise, informal patterns, and non-standard spellings. The model performed well despite these inconsistencies, but the 25% drop between validation and test accuracy shows the sensitivity of sentiment models to linguistic noise.

4. Experimental enhancements provided modest but meaningful improvements

Increasing max token length and R-Drop helped the model generalise better, but no configuration surpassed the main model's validation accuracy. This reinforces that the XLM-RoBERTa Base architecture remains the best fit for the dataset.

5. Confusion patterns show predictable behaviour

Negative and neutral classes were frequently confused—likely because Hinglish users often use mild negativity casually, blurring the distinction. Positive sentiment was the easiest for the model to recognise.

5.4 Limitations of the System

Despite the system's strong performance, certain limitations must be acknowledged:

- **Code-mixed spelling variations** are nearly infinite; normalisation methods were not extensively applied.
- **Emoji and slang handling** can be improved, especially because Hinglish social media posts heavily rely on expressive symbols.
- **Sarcasm and humour**—frequent in Indian digital communication—remain difficult to classify using standard transformer architectures.
- **The dataset covers limited domains**, so the model might behave unpredictably on topics outside the training distribution.
- **Resource constraints** prevented deeper experiments (e.g., longer training, data augmentation, ensemble models, etc.).

Recognising these limitations creates a foundation for future research directions.

5.5 Future Work

The project opens several promising avenues for further enhancement:

1. Data Augmentation

Techniques such as back-translation, synonym replacement, or generative augmentation using LLMs could diversify the training set and improve robustness.

2. Better Handling of Hinglish Orthography

Spell correction, normalisation, and phonetic transliteration may help reduce noise and improve model understanding.

3. Domain Adaptation

Fine-tuning the model on domain-specific Hinglish (e.g., political tweets, movie reviews, or customer feedback) could significantly raise real-world accuracy.

4. Ensemble Models

Combining multiple transformer models could improve classification stability and reduce confusion in borderline cases.

5. Sarcasm & Humour Detection

A separate subsystem focusing on sarcasm could be integrated, enhancing accuracy on difficult posts.

6. Deployment as a Real-Time Application

Future work could involve developing an interactive web or mobile application for live sentiment prediction on social media text.

7. Exploring Other Multilingual Models

Models like mT5, IndicBERT, or LLaMA-based multilingual systems might outperform XLM-R under certain configurations.

5.6 Concluding Remarks

Overall, this project successfully demonstrates that multilingual transformer models can perform strongly on Hinglish sentiment analysis despite the inherent complexities of code-mixed text. Through systematic experimentation, error analysis, and evaluation against real-world test data, the study provides valuable insights into both the strengths and limitations of transformer-based sentiment classification in Indian social media contexts.

The findings reaffirm the importance of multilingual pretraining and highlight specific strategies that can further improve performance. As Hinglish continues to grow across digital platforms, the need for robust NLP systems like this will become even more essential. This project contributes to that ongoing development and lays a strong foundation for future research in multilingual and code-mixed natural language processing.

CHAPTER 6

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